

Homework assignment for 02YELMA

Chapter: Electromagnetic waves

1. We know that if electric fields $\vec{E}$ and magnetic fields $\vec{B}$ exist in vacuum (in absence of charges and currents), they behave like waves according to Maxwell's equations. For now, let us focus only on electric fields. Consider that $\vec{E}$ propagates in the z direction and behaves like plane waves given in the form $\vec{E}(z, t) = \vec{E}_0 e^{i(kz - \omega t)}$ where the phase δ is absorbed in the constant $\vec{E}_0 \equiv \vec{E}_0 e^{i\delta}$ and $\vec{E}_0$ is the real amplitude of the electric field. Note that the real amplitude can be written in general (in 3 dimensions) as $\vec{E}_0 = (E_0)_x \hat{x} + (E_0)_y \hat{y} + (E_0)_z \hat{z}$. By considering $\vec{\nabla} \cdot \vec{E} = 0$ (or equivalently, $\vec{\nabla} \cdot \vec{E} = 0$), show that the component of electric field in the propagation direction is zero, i.e., $(E_0)_z = 0$. Is this also valid for magnetic fields?
2. Consider Faraday's law $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$ and $\vec{E}(z, t) = \vec{E}_0 e^{i(kz - \omega t)}$ to show that $(\vec{B}_0)_y(\omega/k) = (\vec{E}_0)_x$ and $(\vec{B}_0)_x(\omega/k) = -(\vec{E}_0)_y$. If we also consider $(E_0)_z = (B_0)_z = 0$, explain how we can summarize these four equations with a single equation:

$$(\vec{B}_0) = \frac{k}{\omega} (\hat{z} \times \vec{E}_0).$$

Using the equation above as guidance, briefly describe how electric and magnetic fields behave in vacuum.

3. A plane electromagnetic wave in vacuum propagates in the $+z$ -direction. Its electric field is given by

$$\vec{E}(z, t) = E_0 \cos(kz - \omega t) \hat{x}$$

where E_0 , k , and ω are positive constants.

- (a) Using Maxwell's equations, determine the corresponding magnetic field $\vec{B}(z, t)$.
- (b) Compute the instantaneous Poynting vector

$$\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B}$$

and show explicitly that it points in the direction of propagation.

- (c) Find the time-averaged Poynting vector $\langle \vec{S} \rangle$.
- (d) Show that the instantaneous electromagnetic energy density

$$u = \frac{1}{2} \epsilon_0 E^2 + \frac{1}{2\mu_0} B^2$$

can be written entirely in terms of the electric field amplitude E_0 , and prove that the electric and magnetic fields contribute equally to the total energy density.

4. A parallel-plate capacitor with circular plates of radius R is being charged by a steady current I . The plate separation is d , with $d \ll R$, so edge effects may be neglected.

At a given instant, the electric field between the plates is uniform and directed along the z -axis:

$$\vec{E}(t) = E(t) \hat{z}.$$

Although no conduction current passes through the vacuum gap between the plates, energy is still transferred into the capacitor.

- (a) Using Maxwell–Ampère’s law, determine the magnetic field $\vec{B}(r)$ between the plates for $r < R$, where r is the distance from the central axis.
- (b) Compute the Poynting vector

$$\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B}$$

in the region between the plates, and determine its direction.

- (c) Show that electromagnetic energy flows into the capacitor from the sides rather than along the wire axis.
- (d) Calculate the total rate of energy flow into the volume between the plates by integrating the Poynting vector over the cylindrical side surface.
- (e) Show that this equals the rate of increase of electrostatic energy stored in the capacitor:

$$U = \frac{1}{2} \varepsilon_0 E^2 (\pi R^2 d).$$

- (f) Explain conceptually why this example is important for understanding displacement current and the physical meaning of the Poynting vector.

5. A plane electromagnetic wave propagates in vacuum in the $+z$ -direction. Its electric field is given by

$$\vec{E}(z, t) = \hat{x} E_0 \cos(kz - \omega t) + \hat{y} E_0 \cos(kz - \omega t + \phi),$$

where E_0 is a real positive constant and ϕ is a constant phase difference.

- (a) Eliminate the time dependence to obtain the trajectory traced by the tip of the electric field vector in the xy -plane at a fixed position $z = z_0$.
- (b) Determine the special values of ϕ for which the wave is:
- linearly polarized,
 - circularly polarized.
- (c) For the circular polarization case, determine whether the rotation is clockwise or counterclockwise when viewed looking in the $+z$ -direction.
- (d) Using the corresponding magnetic field, verify that the time-averaged Poynting vector points along the direction of propagation and explain why its magnitude is independent of ϕ even though the polarization state changes.